



Project Scope

Interactive System for Macaw Welfare

Planète Sauvage - Port-Saint-Père

Macaw Project Team

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October 2025

Abstract

This document defines the complete scope of the interactive system project designed to improve the welfare of macaws at Planète Sauvage. It covers the geographic, business, functional, technical and budgetary aspects of the project, clearly identifying the boundaries and limits of the intervention. For each dimension, a comprehensive analysis is presented including the existing situation, needs, state of the art, target, and trajectory.



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1 Scope Overview

The project aims to develop an innovative interactive system to cognitively and socially stimulate captive macaws. The scope is defined according to five key dimensions, illustrated below:

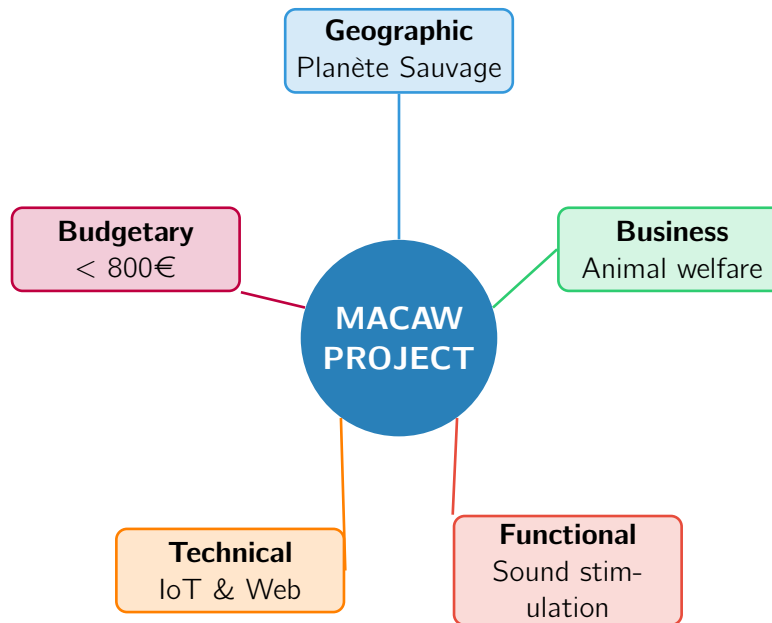


Figure 1: The five dimensions of the project scope

2 Geographic Scope

2.1 Current Situation Analysis

Current Planète Sauvage infrastructure:

- A small island reserved for macaws (figure 2)
- Area: Approximately 120m² for the island
- Presence of a perch where birds often land during the day (figure 3)
- Power supply: Electrical access points at reasonable distance, macaws already have an electric heating device
- Existing technical room with possibility to host server



Figure 2: Macaw island



Figure 3: Macaw perch

2.2 Needs Analysis

The system must be able to:

- Cover the entire macaw island
- Function outdoors with weather protection
- Communicate with a central server located in a technical room
- Be accessible for maintenance by the park's technical team

2.3 Target Analysis

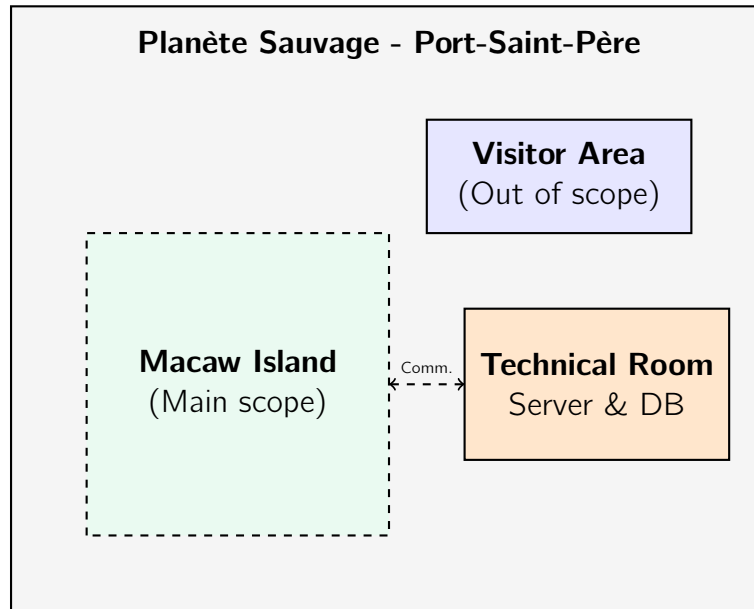


Figure 4: Geographic delimitation of the system

Identified geographic constraints:

- Outdoor installation requiring IP65 minimum waterproofing
- Avoid public spaces for visitors
- Accessibility for technical maintenance

2.4 Trajectory Definition

Phase 1: Installation and materials selection

Phase 2: First prototype installation in the enclosure

Phase 3: Prototype improvement

3 Business Scope

3.1 Current Situation Analysis

Current enrichment practices at Planète Sauvage: Currently there are no existing distractions for macaws, they have very monotonous days.

3.2 Needs Analysis

The macaws' needs are as follows:

- Reduce stereotypical behaviors
- Increase daily cognitive stimulation
- Promote social interactions between individuals
- Have objective behavioral data
- Maintain cognitive activity during periods without caretakers

3.3 Target Analysis

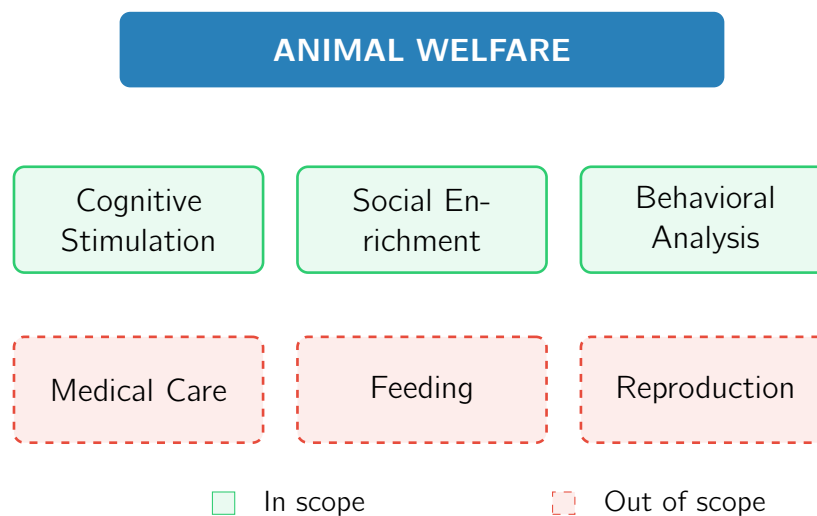


Figure 5: Project business scope

Business objectives:

1. Automate cognitive stimulation of macaws
2. Provide caretakers with information on interactions between the system and macaws

4 Functional Scope

4.1 Current Situation Analysis

Systems currently in place:

- No automated enrichment system
- Manual observations by caretakers
- Absence of behavioral data

4.2 Needs Analysis

User functional needs:

- **Caretakers:** Simple interface, automatic reports, alerts in case of anomaly
- **Veterinarians:** Access to behavioral data
- **Researchers:** Access to behavioral data and export for analysis

4.3 State of the Art

Existing solutions studied:

- **BeakBox:** Commercialized system for domestic macaw playing music in response to physical interaction with the macaw's beak
- **Decision:** Development of custom solution adapted to constraints.

4.4 Target Analysis - Spider Diagram

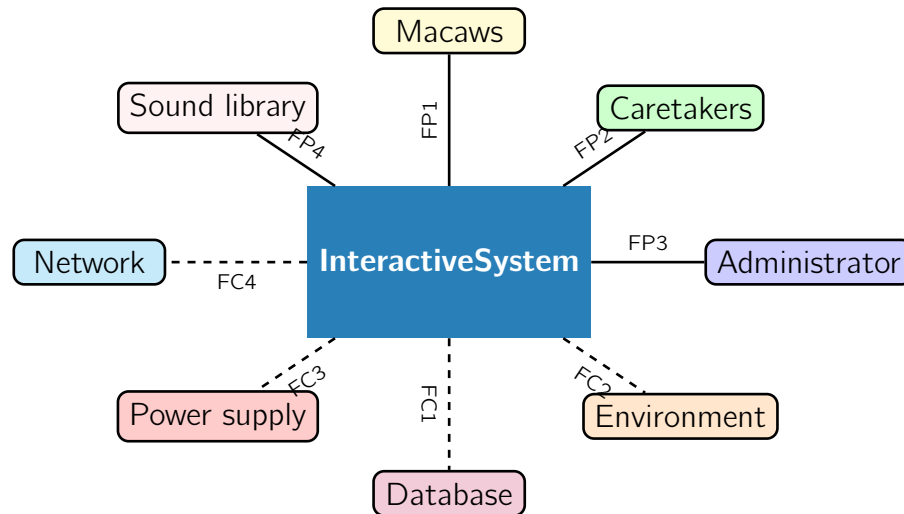


Figure 6: System spider diagram

Primary Functions (FP)

- **FP1:** Stimulate
- **FP2:** Inform
- **FP3:** Configure
- **FP4:** Broadcast

Constraint Functions (FC)

- **FC1:** Store
- **FC2:** Resist
- **FC3:** Power
- **FC4:** Communicate

4.5 Included and Excluded Functions

In scope	Out of scope
✓ Scheduled sound broadcast	× Video broadcast
✓ Reaction recording	× Camera tracking
✓ Basic statistical analysis	× AI behavior recognition
✓ Web supervision interface	× Mobile application
✓ Data export (CSV)	× Park ERP integration
✓ Email alerts	× Push notifications
✓ Manual/automatic mode	× Voice control

Table 1: Functional delimitation

4.6 Functional Trajectory Definition

Version 1.0: Sound broadcast in response to macaw's physical interaction

Version 2.0: Enrichment and analysis of macaw usage statistics

Version 3.0: More advanced data analysis and ability to remotely change sounds produced by the system

5 Technical Scope

5.1 Current Situation Analysis

Available technical infrastructure (To be determined with Planète Sauvage):

- Electrical access
- WiFi network
- Access to existing local servers
- Internal skills

5.2 Needs Analysis

Identified technical needs:

- Low-cost and easily maintainable solution
- Compatibility with existing infrastructure
- Preferred open-source technologies
- Scalability for future extension

5.3 State of the Art

Evaluated technologies:

Technology	Advantages	Disadvantages
Arduino	Very low-cost	Limited power
ESP32	Integrated WiFi, economical	Programming complexity
Raspberry Pi	Versatile, Linux	More expensive
Cloud solutions	Scalable	Dependency, recurring costs

Table 2: Hardware solutions benchmark

Justified choice: Raspberry Pi 4 for its versatility and mature ecosystem.

5.4 Target Analysis - Architecture

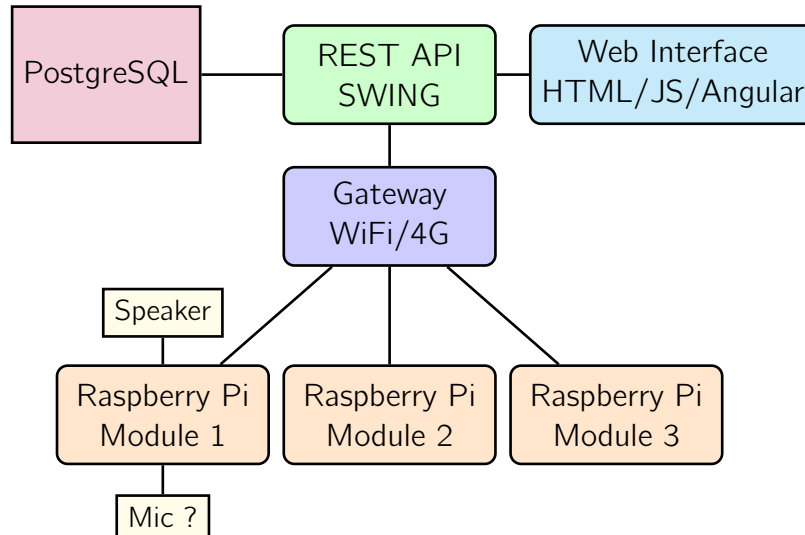


Figure 7: System technical architecture

5.5 Technology Stack

Component	Selected technologies	Discarded alternatives
Hardware	Raspberry Pi 4B (4GB RAM)	Arduino, ESP32
OS	Raspberry Pi OS (Debian)	Ubuntu Server
Backend	Python (Flask/FastAPI)	Node.js, Java Spring
Database	PostgreSQL	MySQL, MongoDB
Frontend	HTML5/CSS3/JavaScript	Native application
Communication	REST API + MQTT	GraphQL, WebSocket
Monitoring	Logs + Simple dashboard	Grafana, Prometheus

Table 3: Technology choices

5.6 Minimum Technical Specifications

- **Performance:** Latency < 200ms for sound broadcast
- **Capacity:** Minimum 1000 events/day
- **Storage:** 32 GB local, expandable
- **Availability:** 99% during trial period
- **Security:** HTTPS encryption, basic authentication



5.7 Technical Trajectory Definition

Phase 1: Development and laboratory testing (functional prototype)

Phase 2: Pilot module deployment on site with intensive monitoring

Phase 3: Industrialization and multi-module deployment

6 Budgetary Scope

6.1 Current Situation Analysis - Current Budget

Previous investments in animal enrichment:

- No existing project for macaws
- Reusable material: local server?, WiFi network

6.2 Budgetary Needs Analysis

Project budgetary constraints:

- Total allocated budget: approximately 700€
- Funding: Park's R&D budget
- No recurring budget first year

6.3 State of the Art - Budget Benchmark of Other Existing Animal Monitoring Solutions

Solution	Description	Cost
ZooMonitor Pro	Complete commercial system	12000€
AnimalTracker	Sensors + software	8000€
DIY Arduino	Basic homemade solution	200€
DIY Raspberry Pi	Advanced open-source solution	600€
Our project	Prototype 1 module	700€

Table 4: Budget positioning

6.4 Cost Breakdown

Category / Item	Quantity	Unit price	Total
Hardware (Prototype phase - 1 module)			
Raspberry Pi 4B (4GB)	1	80€	80€
64GB microSD card	1	20€	20€
Waterproof amplified speaker	1	45€	45€
USB microphone	1	25€	25€
IP65 protection enclosure	1	30€	30€
Power supply + cables	1	15€	15€
Hardware Subtotal			215€
Infrastructure			
Cloud hosting (6 months)	1	60€	60€
Domain name (.fr)	1	12€	12€
Infrastructure Subtotal			72€
Software and licenses			
Licenses (Open Source)	-	0€	0€
Development tools	-	0€	0€
Software Subtotal			0€
Testing and validation			
Testing equipment	-	50€	50€
Consumables	-	30€	30€
Testing Subtotal			80€
Contingency (10%)			
Unexpected costs	-	50€	50€
GRAND TOTAL			417€

Table 5: Detailed project budget (optimized for 1 module)

Available margin: 283€ (for adjustments or improvements)

6.5 Human Resources Budget

Resource	Time	Cost	Covered by
Project student (development)	200h	0€	Academic internship
Technical supervisor	20h	0€	Included supervision
Reference caretaker (testing)	15h	0€	Work time
IT technician	10h	0€	Internal resources
HR Total	245h	0€	

Table 6: Mobilized human resources

6.6 Budgetary Trajectory Definition

Budget phases:

- **Phase 1** (Months 1-3): 500€ - Prototype 1 module
- **Phase 2** (Months 4-6): 500€ additional (if validated) - Module 2
- **Phase 3** (Months 7-9): 500€ additional - Module 3
- **Maintenance** (Year 2+): 185€/year

7 Conclusion

This document establishes a clear, structured scope that meets academic requirements for the interactive system project for macaws at Planète Sauvage.

Key scope points:

- **Controlled budget:** 417€ planned out of 700€ allocated
- **Open technologies:** Open-source solutions for sustainability
- **Scalability:** Architecture allowing future extension

Methodological compliance:

For each of the 5 scope dimensions (geographic, business, functional, technical, budgetary), this document presented:

1. A factual analysis of the current situation
2. An open analysis of needs
3. A documented state of the art
4. A clear definition of the target
5. A realistic trajectory to reach it

The descriptions are adapted to two levels: a communication version for decision-makers and stakeholders, and a detailed technical version for project experts.

The project's success will depend on strict adherence to this scope during the initial phase, with possibility for progressive extension based on feedback and measured results on animal welfare.

8 Bibliography

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